

DUBAI BIM Standard

Version 1.4



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Part 1 – INTRODUCTION AND GENERAL REQUIREMENTS

1. INTRODUCTION

1.1. CONTEXT

The main Building Permit Authorities in Dubai (hereinafter referred to as the Building Permit Authorities) collaborate to deliver a more efficient, transparent, and integrated building permit system. This collaboration aims to accelerate permit approvals, improve building design quality, improve customer satisfaction and enhance Dubai's competitiveness and investment environment. The strategy is based on three pillars:

- Streamlining the building permit application and approval process
- Enabling automated regulatory compliance and validation
- Supporting the transition to a fully digital construction ecosystem

Building Information Modelling (BIM) is adopted as a key enabler to support this strategy enhancing efficiency across the building permit lifecycle.

While this document refers broadly to the Building Permit Authorities in Dubai, it should be noted that, as of the date of issuance, BIM submission is mandated only by Dubai Municipality through its applicable circulars. The terminology used in this standard reflects the coordinated regulatory framework and anticipates future adoption of BIM requirements by other Building Permit Authorities in Dubai.

1.2. PURPOSE

The purpose of this document is to define standardized BIM exchange information requirements for building permit submissions in Dubai, aligned with the principles of ISO 19650:2018, ISO 7817-1:2024, and ISO 16739-1:2024. The standard supports the Dubai BIM Roadmap (2020–2023) and put in operation previous Dubai Municipality BIM circulars. BIM submission has been mandated for specific project types from 1 January 2024 and extended through subsequent circulars (Dubai Municipality, Circular 9-1-3, 2026).

As the receiving party, the Building Permit Authorities in Dubai require consistent, structured, and reusable digital information to enable efficient permit review and downstream use.

1.3. SCOPE

This standard defines the minimum BIM exchange information requirements for submissions to the Building Permit Authorities, in accordance with ISO 19650-1 and ISO 19650-2.

The standard applies exclusively to information submitted for the purpose of obtaining building permits at any stage of the project's lifecycle. Where conflicts arise between this document and appointing party requirements, the more detailed or stringent requirement shall apply. This document does not supersede contractual obligations.

1.4. INFORMATION PURPOSE / INTENDED USE

All BIM models submitted to the Building Permit Authorities shall be prepared to satisfy defined regulatory information purposes. Submitted information shall be complete, consistent, and suitable for its intended use throughout the building permit and inspection processes.

The Information Purpose for BIM submission includes but is not limited to:

- 3D visualization & review of federated building design models
- Automated code compliance checking in accordance with the Dubai Building Code
- Geographic Information System (GIS) and Dubai Digital Twin integration
- Building permit technical assessment and decision-making
- Support Control and Inspection services during the construction and completion phases.

All information requirements defined in this standard shall be interpreted and applied in relation to these Information Purposes.

2. GLOSSARY

Term / Acronym	Definition
3D model	Graphical Data communicating the facility/assets height, width, and depth (X, Y and Z coordinates) characteristics
Activities	Activities are general tasks that need to be carried as part of accomplishing Critical Success Factors. Each activity or groups of activity shall have a specific target against which performance may be measured.
AEC	Architecture, Engineering and Construction Industry
Asset	Something that provides use or value, generally made up of components / assemblies and/or systems
BEP; BIM Execution Plan	The BEP is the document produced by the lead appointed party(ies) to set out their methodology in compliance with the appointed parties information requirements.

BIM; Building Information Modelling	BIM is a process for creating and managing information on a construction project throughout its whole life cycle. In ISO 19650 part 1, BIM refers to the "use of a shared digital representation of a built asset to facilitate design, construction and operation processes to form a reliable basis for decisions"
Building Card	A building card consists of key project information and is usually filled by the applicant upon permit application (usually filled via online portal). Typical information hosted in the building card can be the building name, type, height, and space information like usages and areas.
BCF	An open standard file format used to communicate model-based issues, comments, and viewpoints between project stakeholders without exchanging full model files.
CDE	Common Data Environment is a central information repository that can be accessed by all stakeholders in a project.
Classification	A standardised structure for categorising and indexing data and information
Coordination	The linking of graphical information models to ensure there are no conflicts within the project information model
Cost estimation	Enabling an information model to produce a bill of quantities and/or cost estimation
Data	Set of values, stored but not yet interpreted or analysed, in a form that is convenient to move or process
Database	Collection of data that is organised so that its contents can easily be accessed, managed, and updated
Design authoring	Graphical representation of a design in 3-dimensional space with associated non-graphical information linked
Design reviews	Use of the information models, particularly graphical, to review the progress and technical content of a design
DMD Level	DMD Level refers to the official Dubai Municipality Datum (reference elevation level) used as the base height reference for all construction projects in Dubai. It acts like the "zero height reference" for the entire emirate.

e-Submission (Model)	A model that complies with all the requirements for BIM e-submission and is submitted to respective regulatory agencies.
Federated Model	A combined Building Information Model that has been compiled by linking or combining several different models into one
GIS: Geographic information system	A Geographic Information System (GIS) is a computer system for capturing, storing, checking, and displaying data related to positions on Earth's surface.
IDS	A machine-readable specification used to define and validate the required information (geometry, properties, classifications) that must be delivered in BIM models for a specific information exchange or use.
IFC	Industry Foundation Classes are a set of standardized, digital descriptions of the built asset industry. It is an open, global standard published under a Creative Commons license, and as ISO 16739. IFC provides machine interpretability of information and thereby enables automation of workflows. It is vendor-neutral and available to everyone.
Information	Data which has been interpreted and processed (such as formatting and printing) to take on meaning in some context for its intended receiver
Information Model	Information models form the cornerstone of all digitisations of AEC information as it is the repository of data that are required for all intended uses and analyses. As such, the validity of the data inside information models should be dictated by its suitability for use.
LOIN	A structured definition of the quality, quantity, and granularity of information required at a specific project stage and for a specific purpose, including geometrical and alphanumerical information, in accordance with ISO 19650 and ISO 17412-1.
Model	Generic term for a virtual representation of an asset
Project information model	A digital representation of the Project or part of the Project, and used to describe a two-dimensional representation, three-dimensional representation, as well as other data representations
Schema	A representation of a plan in the form of an outline structure

Spaces / rooms	In this instance, either a space or a room is a defined area within a model file that has attributes and boundaries providing a basis for compliance checking.
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3. CONCEPTS AND PRINCIPLES

3.1. BUILDING INFORMATION MODELLING (BIM)

BIM is defined as the use of a shared digital representation of a built asset to support design, construction, and operation processes, forming a reliable basis for decision-making (ISO 19650-1).

For building permits in Dubai, BIM is further defined as the digitisation of construction information to improve quality, efficiency, and reusability throughout the permit lifecycle (Roadmap for BIM implementation, 2020).

3.2. INFORMATION MANAGEMENT

Information and how it is exchanged and managed is a crucial aspect of implementing and using BIM successfully. All parties submitting information models are expected to:

- Understand ISO 19650-1 concepts and principles
- Apply ISO 19650-2 delivery processes
- Use this document as a Dubai-specific addendum where no national annex exists

3.2.1. RELATIONSHIPS BETWEEN PARTIES

The Building Permit Authorities do not have contractual relationships with the parties involved in the production of project information. Their role is defined solely as a Receiving Party. For clarity, the roles applicable to the building permit process are defined as follows:

- **Receiving Party:** The relevant Building Permit Authorities reviewing and authorising submitted information models.
- **Appointing Party:** The client, such as a developer, landowner, or government entity, responsible for appointing the delivery team.
- **Lead Appointed Party:** The lead consultant or contractor responsible for coordinating information delivery.
- **Appointed Party:** Consultants or contractors appointed by the Lead Appointed Party.

These roles are presented to align the building permit process with ISO 19650 information management principles and do not alter contractual responsibilities between project parties.

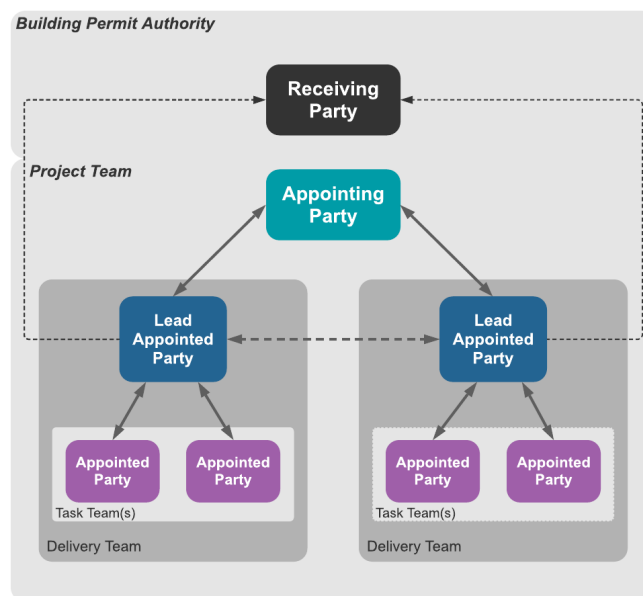


Figure 1 - Project Team hierarchy as per ISO 19650

3.2.2. INFORMATION MANAGEMENT PROCESS

The information management process for BIM submissions to the Building Permit Authorities shall follow the principles and activities defined in ISO 19650-2:2018. Due to the absence of a contractual relationship between the Building Permit Authorities and the project delivery team, this section clarifies the points within the ISO 19650-2 process where interaction with the Receiving Party (The Building Permit Authorities) occurs in relation to building permit submissions.

The information exchange process comprises eight activities as defined in ISO 19650-2. Only those activities that directly affect BIM submissions to the Building Permit Authorities are addressed/highlighted in this document.

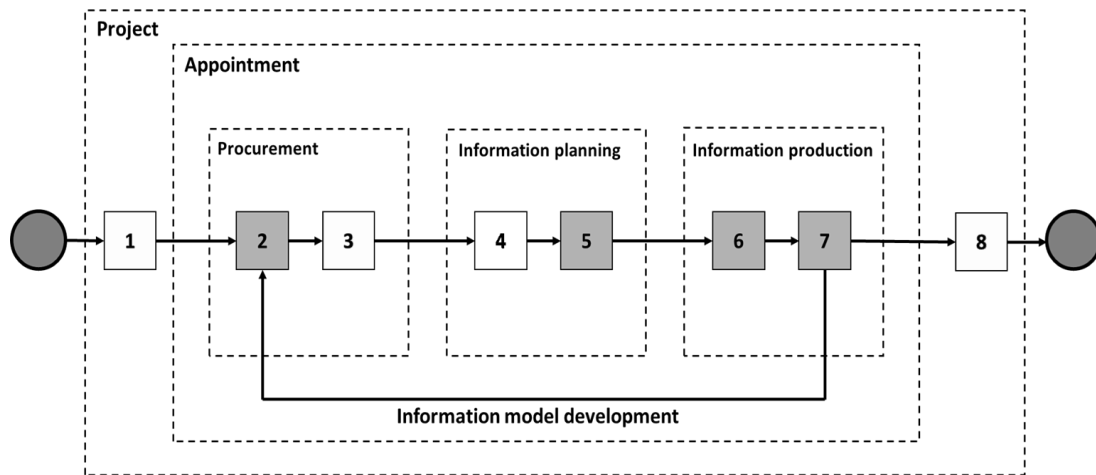


Figure 2 – Information management process as per ISO19650-2 with highlighted stages appropriate to this document.

Key to Figure: 1 - Assessment and need, 2 – Invitation to tender, 3 – Tender response, 4 – Appointment, 5 – Mobilization, 6 – Collaborative production of information, 7 – Information model delivery, 8 – Project close-out (end of delivery phase)

Where BIM submission is mandated by the Building Permit Authorities, the actions described in this section shall be applied. For projects where BIM is not mandated, application of this process is still recommended.

The responsibilities of the Appointing Party, Lead Appointed Party, and Appointed Parties in relation to building permit submissions are summarised in Table 1. These responsibilities relate solely to information delivery for permitting purposes and do not alter contractual obligations between project parties.

BIM Process (ISO 19650-2)			Responsible Party	Action Related to Permit Submission	Authority Requirement
2.	Invitation to tender	Establish the information requirements including information standards, production methods and procedures and protocol	Appointing Party	Append this BIM standard to information requirements	Recommended

4.	Appointment	Confirm the appointed parties' approach and documents pertaining to information management	Lead/Appointed Parties	Confirm approach to comply with Authority requirements	Recommended
5.	Mobilisation	Preparation for delivery of the project	Lead/Appointed Parties	Prepare information containers per this document	Mandatory (where BIM mandated)
6.	Collaborative production of information	Development of the project information model	Lead/Appointed Parties	Develop information required for permit submission	Mandatory
7.	Information model delivery	Submission of project information model for Appointing Party authorisation	Lead/Appointed Parties	Submit validated information model for authority review	Mandatory

Table 1 – Information workflow stages and associated deliverables as per ISO 19650

3.2.3. BIM EXECUTION PLAN (BEP)

There is no requirement for a BIM Execution Plan (BEP) to be submitted to, reviewed by, or approved by the Building Permit Authorities as part of the building permit process. The preparation, content, and management of the BEP remain the responsibility of the Appointing Party and the project delivery team in accordance with their contractual arrangements.

Where a BEP is used, it is recommended that appointed parties incorporate the requirements of this document into their project, which should be reviewed internally by the Appointing Party or their appointed information manager.

This recommendation is intended to support efficient information delivery and reduce the risk of non-compliance at the time of BIM submission. The absence or content of a BEP shall not, in itself, affect building permit assessment by the Building Permit Authorities.

4. GENERAL EXCHANGE INFORMATION REQUIREMENTS

This section defines the general exchange information requirements for the submission of BIM information models to the Building Permit Authorities. These requirements apply to all BIM submissions made for the purpose of obtaining building permits, at any stage of the project lifecycle, where BIM submission is mandated by the Building Permit Authorities.

The requirements set out in this section establish the minimum conditions for the preparation, organisation, and delivery of information models intended for regulatory review. Detailed technical requirements are defined in Part 2 – Exchange Information Requirements of this document.

All parties submitting information models to the Building Permit Authorities shall ensure that the information delivered is in accordance with the Information Purpose defined in Section 1.4.

Where BIM submission is not mandated, application of these requirements is recommended to support consistency and future digital integration.

4.1. INFORMATION CONTAINER REQUIREMENTS AND EXCHANGE FORMATS

The Building Permit Authorities act as the Receiving Party for information submitted in support of building permit applications. All information models submitted for regulatory review shall be delivered as defined information containers using openBIM standards.

4.1.1. INFORMATION CONTAINERS

All primary BIM information models submitted for building permit purposes shall be provided in Industry Foundation Classes (IFC) schema. IFC models constitute the authoritative information container for permit assessment and to support their intended use, in accordance with the Information Purpose defined in Section 1.4. The use of openBIM ensures vendor-neutral information exchange and interoperability between BIM authoring tools and the Authorities' review systems.

To support consistent and correct information exchange, the Authorities may provide templates and guidance for commonly used BIM authoring platforms, including recommended mappings to the appropriate Model View Definitions (MVDs). Use of these templates is recommended but does not replace the requirement to comply with this standard.

Additional requirements relating to file formats, file size limitations, naming conventions, schema versions, and model structuring are defined in subsequent sections of **Part 2 – Exchange Information Requirements**.

4.1.2. GEO-REFERENCING

All information models submitted to the Building Permit Authorities for the purpose of obtaining building permits shall be geo-referenced at 1:1 scale to real-world coordinates. Geo-referencing shall be established through alignment between a permanent geographical coordinate reference system and the local coordinate system used within the information model.

All submitted IFC files shall include easting, northing, and elevation values referenced to the following coordinate reference system:

Coordinate Reference System: EPSG:3997

Description: WGS 84 / Dubai Local Transverse Mercator (TM)

Geo-referencing information shall be embedded within the IFC file at the time of submission and shall support integration with the Building Permit Authorities' Geographic Information Systems (GIS). BIM authoring tools used for the development of information models are expected to support geo-referencing and IFC export of coordinate reference information.

4.2. INFORMATION MODEL

All information models submitted to the Building Permit Authorities shall be prepared to support their intended use, in accordance with the Information Purpose defined in Section 1.4 .

Information models shall comply with the requirements defined in this section and the detailed rules set out in **Part 2 – Exchange Information Requirements**, including Level of Information Need, information modelling, and information containers format.

4.2.1. LEVEL OF INFORMATION NEED

The Level of Information Need defines the minimum geometrical and alphanumerical information required for information models submitted in support of a building permit application, in accordance with ISO 17412-1 and ISO 19650.

The required Level of Information Need shall depend on the type of building permit application and the project stage. The Building Permit Authorities define the minimum information required for each submission to enable verification of compliance with applicable building codes. The Level of Information Need shall be specified using a combination of geometrical and alphanumerical information requirements. Levels may differ between model elements, disciplines, and permit stages.

4.2.2. CLASSIFICATION

All information models submitted to Building Permit Authorities shall apply the classification system specified in this document. Classification shall be based on local usage codes defined by Building Permit Authorities and applied

consistently within the submitted information containers, including building types, units, and spaces. The use of additional or project-specific classification systems may be applied for internal project purposes; however, they shall not replace or conflict with the required local classification codes used for permit review.

5. ROLES AND RESPONSIBILITIES

From the outset of any project implementing BIM, roles and responsibilities related to information model development and submission shall be clearly defined by the project delivery team.

The roles described in this section relate solely to interactions between the Building Permit Authorities, as Receiving Party, and the parties responsible for submitting information models for building permit purposes. These roles do not alter contractual responsibilities between project parties.

- Receiving Party (Building Permit Authority Engineer):
 - Review submitted information models for completeness and compliance with permit requirements.
 - Verify that submitted information models satisfy their intended use, in accordance with the Information Purpose defined in Section 1.4.
 - Provide comments and feedback and authorise compliant submissions.
- Lead Appointed Party / Appointed Party (Information Manager):
 - Ensure submitted information models are complete, quality-checked, and compliant with this standard.
 - Coordinate and issue information models to the Building Permit Authorities.
- Lead Appointed Party / Appointed Party (Information Model Author):
 - Ensure model geometry, attributes, classification, coordinates, and naming conventions comply with the required Level of Information Need and this standard.
 - Ensure correct use of authoring tool templates and IFC export configurations.

Part 2 – EXCHANGE INFORMATION REQUIREMENTS

6. INFORMATION CONTAINER REQUIREMENTS AND EXCHANGE FORMATS

6.1. FILE REQUIREMENTS

6.1.1. FILE FORMAT

All primary BIM files submitted to permitting authorities for the purpose of obtaining building permits shall be in Industry Foundation Classes (IFC) schema. All submitted IFC files shall:

- Be exported using the IFC4 schema and delivered using the .ifc file extension.
- Contain both geometric and alphanumerical information in accordance with the graphical and information requirements defined in this standard.
- Be suitable to support the intended use, in accordance with the Information Purpose defined in Section 1.4.

Where BIM submission is mandated, any file submission requirements that apply to non-BIM submissions shall continue to apply without change. This includes the submission of 2D drawings, which should be generated directly from the source BIM models.

Native BIM authoring files (e.g. proprietary software formats) may be requested by the Building Permit Authorities as supplementary information where additional clarification is required. Where provided, native files shall be treated as supporting documents only and shall not be accepted as a standalone deliverable. In all cases, the IFC file shall be considered the authoritative source of information for regulatory assessment.

6.1.2. FILE SIZE

BIM file sizes shall be managed to ensure efficient performance for both applicants and the Building Permit Authorities, including practical upload and download times. Project-specific BIM file size limits may be defined within the project BIM Execution Plan (BEP) and shall be applied by the project team where relevant. However, for BIM submissions to the permitting authorities, the maximum file size of the ZIP package containing the IFC files shall not exceed the maximum size permitted by the submission platform.

Where this limit cannot be met, applicants are required to coordinate with the Building Permit Authorities to determine an appropriate approach for handling large file submissions.

6.1.3. FILE NAMING

All information models submitted to the Building Permit Authorities for building permit purposes shall comply with the naming fields and formats defined in Table 2. While project-specific file naming conventions may be established within the project BIM Execution Plan (BEP), such conventions shall not override or replace the naming requirements specified by the Building Permit Authorities for building permit submissions.

The naming convention defined below shall be used for all BIM submissions. The naming convention consists of 5 individual fields with an underscore character separating each field.

Project Number	-	Building ID	-	Parcel ID	-	Model Discipline	-	Model Number
PN123456	_	BI123456	_	PA1234567	_	AR	_	001

Field	Description	Character format
Project Number	Project identification number generated and issued by the permitting authority.	Abbreviation "PN" followed by a unique 6-digit number
Building ID	Building identification number used to identify each individual building within a project. Building IDs are sequential, starting from 000001.	Abbreviation "BI" followed by a unique 6-digit number.
Parcel ID	Land parcel identification number predefined by the permitting authority. This is typically included in the Affection Plan data issued by the permitting authority.	Abbreviation "PA" followed by unique 7 or 8 -digit number.
Model Discipline	Identifies the model discipline, e.g., Architecture, Structural, MEP etc.	2-character abbreviation of model discipline (refer to Table 3 - Model discipline abbreviations)
Model Number	Model part identification used when an IFC model is divided into multiple files due to size or complexity constraints. Numbering shall start from 001.	3-digit numerical field.

Table 2 – Naming convention field descriptions

ID	Discipline
AR	Architecture
ST	Structural

Table 3 - Model discipline abbreviations

7. INFORMATION MODEL

7.1. LEVEL OF INFORMATION NEED

The Level of Information Need defines the minimum geometrical and alphanumeric information required within Information Models submitted to the Building Permit Authorities for the purpose of building permit assessment, in accordance with ISO 19650 and ISO 17412-1.

For building permit submissions, the Level of Information Need is applied to ensure that Information Models contain sufficient and appropriate information to support the intended use, in accordance with the Information Purpose defined in Section 1.4. It does not represent the full level of detail required for construction or fabrication purposes.

The required Level of Information Need shall be defined using a combination of geometrical and alphanumerical information requirements. These requirements may vary depending on:

- The type of building permit application.
- The project stage at which the submission is made.
- The model discipline.
- The specific model elements included within the Information Model.

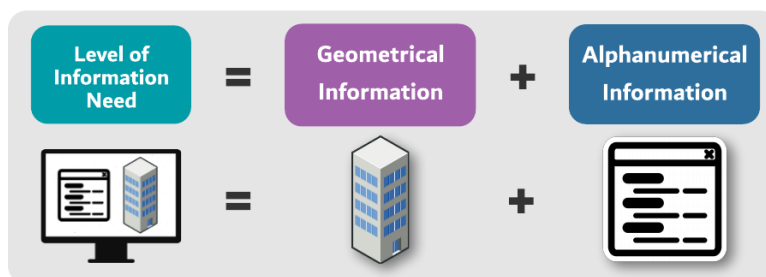


Figure 3 – Level of Information Need

The Level of Information Need shall be applied at the model element level and shall not be interpreted as a single level applicable to the Information Model as a whole. Different model elements within the same submission may therefore be subject to different information requirements. Levels of Information Need have no meaning unless accompanied by defined geometrical and alphanumerical requirements. The detailed information requirements for each model element and permit stage are specified in the supporting appendices to this standard, including:

- Appendix A – Model Element Matrix.
- Appendix B – Model Attribute Matrix.

All Information Models submitted to the Building Permit Authorities shall meet the required Level of Information Need applicable to the relevant permit stage.

Information that is below or beyond the required Level of Information Need may negatively impact the intended use, in accordance with the Information Purpose defined in Section 1.4, and may result in the submission being returned for correction.

7.1.1. GEOMETRICAL INFORMATION

Geometrical information refers to the graphical representation of model elements within an Information Model, including their shape, size, spatial location, and orientation.

For building permit submissions, geometrical information shall be provided at a level sufficient to support regulatory review, automated code compliance checking, and spatial verification. The required geometrical information may vary depending on the permit type, submission stage, model discipline, and specific model elements.

The Level of Information Need for geometrical information shall be satisfied through defined geometrical requirements applied at the model element level. These requirements represent the minimum geometrical information necessary for permit assessment and do not indicate the level of detail required for construction or fabrication purposes. Detailed geometrical requirements for each model element and permit stage are defined in Appendix A – Model Element Matrix.

7.1.2. ALPHANUMERICAL INFORMATION

Alphanumerical information refers to all non-graphical information associated with model elements within an Information Model, including attributes, properties, classifications, identifiers, and usage data.

For building permit submissions, alphanumerical information shall be provided to support their intended use, in accordance with the Information Purpose defined in Section 1.4. Alphanumerical information is a mandatory component of the Level of Information Need and shall be provided in addition to geometrical information.

The required alphanumerical information may vary depending on the permit type, submission stage, model discipline, and specific model elements. Alphanumerical requirements shall be applied at the model element level and mapped to the appropriate IFC classes and property sets. Detailed alphanumerical information requirements are defined in Appendix B – Model Attribute Matrix.

7.2. MODEL ELEMENT REQUIREMENTS

This section defines the general requirements for modelling elements within Information Models submitted to the Building Permit Authorities.

All model elements included in Information Models shall be modelled, categorised, and classified in a manner that supports regulatory review, automated code compliance checking, and interoperability using openBIM standards. Model elements shall be developed in accordance with the Level of Information Need defined in Section 7.1 and the requirements set out in the supporting appendices, as applicable to the relevant permit stage.

Model elements shall be assigned to the appropriate IFC classes, predefined types, and parameters upon export. Incorrect classification or misuse of elements may prevent automated processing and regulatory assessment.

Information that is below or beyond the required Level of Information Need shall be avoided, as it may negatively impact the intended use. The Level of Information Need defined in this standard does not represent construction or fabrication-level modelling requirements.

Detailed requirements for model element inclusion, geometrical definition, and alphanumerical attributes applicable to each permit stage are defined in:

- Appendix A – Model Element Matrix
- Appendix B – Model Attribute Matrix

7.2.1. IFC CLASSIFICATION

All model elements included in Information Models submitted to the Building Permit Authorities shall be classified according to their functional behaviour using the appropriate Industry Foundation Classes (IFC) entities.

To ensure interoperability and enable automated regulatory assessment, all model elements shall be assigned:

- The correct IFC class.
- The appropriate PredefinedType value when defined in the IFC schema.
- An ObjectTypeOverride value where:
 1. The required type is not available as a standard PredefinedType, or
 2. Additional differentiation is required beyond the standard PredefinedType.

The combination of IFC class, PredefinedType, and ObjectTypeOverride shall unambiguously describe the function of each model element for regulatory review and automated processing. Incorrect or inconsistent classification may prevent automated assessment and may result in the submission being returned for correction. Examples of IFC class assignment, PredefinedType usage, and ObjectTypeOverride application are provided in Table 4 for reference.

Object (Authoring tool)	IFC Class (IfcElement)	Predefined Type	Object type override
Ceiling	IfcCovering	CEILING	–
Room/Space	IfcSpace	–	SINGLELOADEDCORRIDOR
Parking bay	IfcSpace	PARKING	PARALLEL

Table 4 - IFC object mapping example

7.3. BUILDING MODELLING

All buildings included in Information Models submitted to the Building Permit Authorities shall be represented using the IfcBuilding entity upon export to IFC. Building-level information required for permit assessment, including identification, classification, and usage data, shall be provided in accordance with the requirements defined in Appendix B – Model Attribute Matrix and Appendix C – Usages Codes.

7.3.1. BUILDING TYPE AND USAGES

All buildings shall be classified according to their building type and usage in accordance with the local codes defined by the Building Permit Authorities.

Building type and usage information is used for regulatory assessment, including verification of zoning compliance, occupancy classification, and calculation of applicable planning parameters. Each building shall be assigned the appropriate building type, master usage, and main usage codes as defined in Appendix C – Usage Codes. Incorrect or missing building type or usage classification may prevent automated processing and regulatory assessment of the submitted Information Model.

Building Type	Master Usages (Occupancy)	Main Usages (Occupancy Use)	Occupancy Usage Code
Multi Storey	RESIDENTIAL	residential apartments	MS_1_1
		labour accommodation	MS_1_2
	COMMERCIAL	shopping centre	MS_3_1
		shops	MS_3_2

Table 5 - Example of building types and usages

7.4. LEVEL NAMING

All building levels included in Information Models submitted to the Building Permit Authorities shall be represented using the IfcBuildingStorey entity upon export to IFC.

Each IfcBuildingStorey shall be clearly named to identify the corresponding building floor or level. Level names shall support efficient model navigation, automated checks, and regulatory assessment and shall be applied consistently across all submitted models. The level naming convention defined in this standard shall be used for all BIM submissions for building permit purposes.

7.4.1. LEVEL NAMING FORMAT

Each physical floor or level within a building shall be represented by a corresponding IfcBuildingStorey and assigned a unique level name using the defined format. The level name shall consist of two mandatory fields separated by an underscore (_):

- Level Abbreviation
- Level Identification

Level Abbreviation	-	Level Identification
F1	_	FLOOR1

The Level Abbreviation field shall identify the level designation and shall use upper-case characters only (e.g. B1, GR, P1, M1, F1, RF). The Level Identification field shall describe the level in accordance with the abbreviation used (e.g. BASEMENT1, Ground Floor, Floor 1, Roof).

The Level Identification field shall describe the level in a manner consistent with the Level Abbreviation. The specific wording of the Level Identification may be defined by the applicant in accordance with their internal project standards, provided that it remains clear, unambiguous, and consistently applied.

Figure 4 illustrates a sample building and its levels.

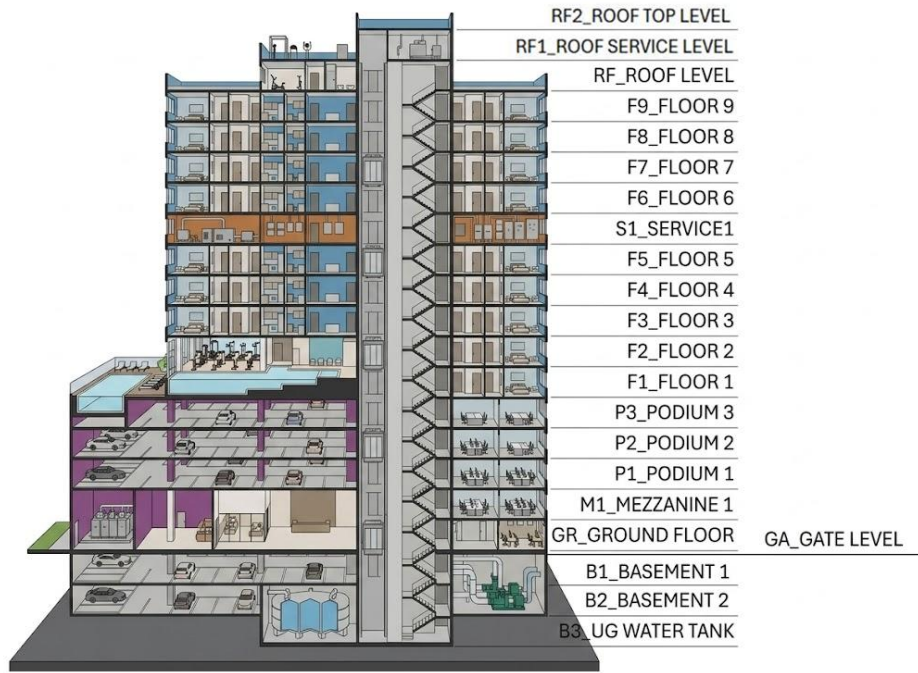


Figure 4 – Building Floor/Level ID

7.5. SPACE/ROOM MODELLING

7.5.1. GENERAL

The inclusion of space/room elements in Information Models submitted to the Building Permit Authorities is mandatory for all BIM submissions made for permit purposes.

Space/room elements and their associated information are critical to enabling automated code compliance checking, spatial analysis, and regulatory verification, including but not limited to the calculation of: Gross Floor Area (GFA), Net Area (NA), Built-Up Area (BUA), Floor Area Ratio (FAR), and Occupancy and space usage compliance.

Spaces/rooms shall be modelled as bounded areas or volumes using the BIM authoring tool's dedicated space definition functionality and exported to IFC using the appropriate IFC entity. Each distinct functional area within the building shall be represented by an individual space/room element.

7.5.2. IFC ENTITY AND CLASSIFICATION

All spaces/rooms shall be exported using the IfcSpace entity. Each IfcSpace shall:

- Be correctly positioned within the corresponding IfcBuildingStorey
- Be geometrically bounded and spatially valid
- Contain the minimum geometrical and alphanumeric information required to meet the applicable Level of Information Need

Incorrect or missing space elements may prevent automated processing and may result in the submission being returned for correction.

7.5.3. SPACE/ROOM NAMING

Each space/room shall have its Name parameter populated. The space/room name shall:

- Clearly describe the functional use of the space
- Be human-readable
- Be applied consistently across the Information Model

Space/room naming is independent from space usage classification and does not replace the requirement for assigning usage codes.

7.5.4. SPACE/ROOM USAGE CLASSIFICATION

All spaces/rooms shall be assigned a space usage code that identifies their functional use in accordance with the local usage codes defined by the Building Permit Authorities. Space usage information shall be exported to IFC using the following properties:

- SpaceUsageCode – the coded identifier of the space usage
- SpaceUsageDescription – a human-readable description of the space usage

Space usage codes are used for regulatory assessment purposes, including occupancy verification and spatial compliance checks. All space/room elements shall be classified using the relevant codes defined in Appendix C – Usage Codes. Missing or incorrect space usage classification may prevent automated processing and regulatory assessment.

The below table shows examples of the space usage code for a few living spaces.

Zone Category	Space Description	Space Category Codes
Living Space	Living Room	SC_25_15_10
	Sitting Room	SC_25_15_15
	Majlis	SC_25_15_20
	Living-Dinning	SC_25_15_25

Table 6 - Example of space usages codes

7.5.5. CONSISTENCY REQUIREMENTS

Space/room modelling, naming, numbering, and classification shall be:

- Consistent across all disciplines and submitted IFC files
- Aligned with building, level, and usage information
- Compliant with the applicable Level of Information Need for the relevant permit stage

7.6. MODEL SEGREGATION

7.6.1. GENERAL

Where an Information Model is divided into multiple IFC files, model segregation shall be applied in a controlled and consistent manner to ensure effective regulatory review, automated processing, and interoperability.

Model segregation shall comply with the requirements set out in this section and shall not compromise the completeness or integrity of information required for the applicable permit stage.

7.6.2. HIGH-LEVEL SEGREGATION PRINCIPLES

Information Models submitted to the Building Permit Authorities shall comply with the following segregation principles.

- **Building-based segregation:** Each IFC file shall relate to one building only. Each building within a project shall be represented by a separate IfcBuilding entity. Information from different buildings shall not be combined within the same IFC file.
- **Discipline-based segregation:** Information from different disciplines shall be segregated into separate IFC files. Architectural and structural information shall not be combined within a single IFC file.
- **Single source of elements:** Each physical element shall exist in only one IFC file. Element duplication across IFC files is not permitted.
- **Logical model partitioning:** Where a building model is divided into multiple IFC files, each file shall represent a logical part of the same building and shall be identifiable using the model part identification defined in the file naming convention.
- **Consistency across segregated models:** All segregated IFC files forming part of a single submission shall be spatially coordinated, use consistent building and level identification, and share the same geo-referencing and coordinate reference system.

Non-compliance with these rules may prevent automated processing and may result in the submission being returned for correction. Where strict compliance with these segregation principles cannot be met, applicants shall coordinate with the Building Permit Authorities to agree an appropriate approach prior to submission.

7.6.3. COMPLEX STRUCTURES COMPRISING MULTIPLE BUILDINGS

In the case of complex structures comprising multiple buildings, detailed modelling rules, including representation as separate IfcBuilding entities, level continuation above shared podiums, and related naming conventions, are defined in the **Technical Guides**. These guides provide illustrated examples and step-by-step instructions for common scenarios to ensure consistent and compliant BIM submissions.

7.7. MODEL GEO-REFERENCING

All IFC models submitted to the Building Permit Authorities shall be geo-located to real-world coordinates using appropriate easting, northing, and DMD level elevation value. Geo-referencing information shall be embedded within the

IFC file at the time of submission and shall support the intended use, in accordance with the Information Purpose defined in Section 1.4.

Failure to comply with these requirements may prevent GIS integration and automated processing and may result in the submission being returned for correction.

Part 3 – DIGITAL SUBMISSION AND VALIDATION FRAMEWORK

Part 3 defines the regulatory framework governing the digital submission, automated validation, and assessment of information models for building permit applications. It does not introduce additional information requirements beyond those defined in Parts 1 and 2 but establishes how compliance with those requirements is verified and managed to support efficient, consistent, and scalable regulatory review.

8. DIGITAL SUBMISSION PLATFORM

All BIM submissions to the Building Permit Authorities shall be made through the designated digital submission platform. The platform serves as the primary interface for:

- Submission of BIM information models and associated documentation
- Automated validation of submitted information
- Communication of review outcomes and feedback

The platform is designed to support structured data exchange, reduce manual intervention, and enable consistent application of regulatory requirements across all submissions.

9. PRE-SUBMISSION VALIDATION (QA/QC)

Prior to submission, applicants shall ensure that all BIM information models have undergone pre-submission validation against the applicable requirements defined in:

- This BIM Standard; and
- The supporting documents issued by the Building Permit Authorities.

Pre-submission validation shall be carried out using tools capable of verifying compliance with the required information structure, classification, geometrical integrity, and alphanumerical completeness applicable to the relevant permit stage.

The objective of pre-submission validation is to:

- Reduce errors and inconsistencies in submitted models
- Improve the efficiency of regulatory review
- Minimise the likelihood of submission rejection

Prior to submission, applicants shall also verify that all 2D CAD drawings submitted for permitting purposes are generated directly from the corresponding BIM Information Models. Pre-submission validation shall include verification of consistency and alignment between the BIM models and the submitted CAD drawings.

Where required by the Building Permit Authorities, applicants shall submit validation reports demonstrating conformity with the applicable requirements.

10. REGULATORY VALIDATION PROCESS

Upon submission through the designated digital submission platform, all BIM information models shall be subject to regulatory validation conducted by the Building Permit Authorities.

Regulatory validation shall include, but is not limited to:

- Verification of file structure, format, and naming compliance
- Validation of geometrical and alphanumerical information against the required Level of Information Need
- Validation of classification, usage codes, and attribute completeness
- Spatial and rule-based compliance checks, where applicable

The regulatory review of submitted 2D CAD drawings shall remain unchanged. Where BIM submission is required, submitted CAD drawings shall be generated directly from the corresponding BIM Information Models, and part of the regulatory review shall include verification of alignment and consistency between the BIM models and the submitted CAD documentation.

11. REGULATORY DECISION OUTCOMES

Following completion of regulatory validation and review, each BIM submission shall be assigned one of the following regulatory decision outcomes:

- **Approved** – The submission complies with the applicable requirements and is approved to proceed within the building permit process.
- **Conditionally Approved** – The submission substantially complies with the applicable requirements but includes minor non-conformities that shall be corrected within the timeframe specified by the Building Permit Authorities
- **Returned/Consultant for Review** – The submission fails to meet the required information, validation, or compliance criteria and shall be revised and resubmitted prior to further assessment.

Detailed feedback, including identified issues and required actions, shall be communicated through the dedicated systems.

Part 4 – SUPPORTING DOCUMENTATION

This BIM Standard establishes the primary requirements and shall be supported by the documents listed below, which provide additional specification and guidance.

12. TECHNICAL GUIDES

The Technical Guides provide detailed technical guidance to permit applicants in the preparation of BIM information models for submission to the Building Permit Authorities. The Technical Guidelines shall be used in conjunction with this BIM Standard and its appendices to ensure consistent, accurate, and compliant BIM submissions.

These documents collectively define the detailed technical, data, and validation requirements necessary to achieve compliance with this standard.

13. INFORMATION DELIVERY SPECIFICATIONS (IDS)

Information Delivery Specification (IDS) documents support the automated validation of information model quality. However, compliance with IDS alone does not guarantee that an information model fully meets the information exchange requirements defined in this standard and its supporting documents.

14. MODEL ELEMENT MATRIX (APPENDIX A)

The Model Element Matrix document defines what elements need to be included in IFC models being submitted to permitting authorities. It identifies permitting stages and to what level of detail elements should be modelled based on the Level of Information Need outlined in the BIM Standard.

15. MODEL ATTRIBUTE MATRIX (APPENDIX B)

The Model Attribute Matrix document defines the information requirements for building elements. The Model Attribute Matrix refers to all elements referred to in the Model Element Matrix. It lists all element attributes (properties) required for each permitting stage, including information on IFC Reference, Property Set, User Input and Data Type.

16. USAGE CODES (APPENDIX C)

The Usage Codes document defines the list of local codes to identify building types and usages, unit usages, and space usages. The codes are machine-readable validation to support the automated validation of information model quality.

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